

USING AI TO ANALYSE THE PREVALENCE OF NUTRITION, HEALTH AND MARKETING CLAIMS ON CHILD-TARGETED ULTRA-PROCESSED FOODS

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INTRODUCTION

Australian children obtain around half their energy from ultra-processed foods (UPFs), this is a significant public health concern as dietary patterns high in UPFs can increase chronic disease risk.(1) UPF packaging uses regulated health and nutrition claims and unregulated marketing claims, particularly on the front-of-pack (FOP) that can influence parents' and children's perceptions of the product's healthiness and their purchasing decisions.(2) However, there is limited evidence on the prevalence of these FOP claims on UPFs marketed to children in the Australian context. The extent of the use of claims on UPFs in Australia remains unclear due to the manual, resource-intensive methods used to collect and analyse FOP claims. Recent advances in artificial intelligence (AI), including optical character recognition (OCR) and large language models (LLMs) mean that FOP claims can be extracted and analysed more efficiently and on a larger scale. However, the accuracy of AI to extract and classify claims on food packages remains unknown.

This proof-of-concept study aimed to develop and test an AI-enabled system for extracting and classifying claims used on child-targeted ultra-processed breakfast cereals and snack bars, and to describe how they are used on UPFs in comparison to non-UPFs.

METHODS

All breakfast cereal and snack bar images and product information from the websites of Coles and Woolworths were extracted in February 2025 and entered into a database. A purpose-built checklist to determine if products were child-targeted was applied, assigning each product a score (0-10) with higher scores reflecting the presence of more child-targeted strategies on the FOP. UPFs were categorised (Yes/No) using the Nova classification system, which utilises the ingredients list. UPFs were identified by the presence of markers in the ingredients list, including additives of cosmetic function (e.g., flavours, colours, emulsifiers) and/or food substances of rare culinary use (e.g., protein isolates, maltodextrin, interesterified oils). The development of the AI-enabled system involved the use of Microsoft Azure Document Intelligence, a text reading software that uses OCR to identify and extract all the claims present on the FOP based on a developed

prompt. A second prompt was developed for use in a LLM to classify each claim according to a FOP taxonomy- a systematic framework for classifying FOP design components, including claims (i.e., nutrition, health or marketing claim). Accuracy of the AI-enabled system was determined by randomly selecting 5% of the sample to be manually cross-checked, ensuring all the claims were extracted and classified correctly. A threshold of accuracy of 90% was set as acceptable. All data were analysed using SPSS. Data analysis for the prevalence of claims in UPF vs non-UPF is underway.

RESULTS AND DISCUSSION

To date, 305 products have been assessed with the child-targeted checklist, with scores ranging from zero to eight. An AI-enabled system for detecting and classifying claims on the products has been developed and is currently being tested for accuracy. Preliminary findings suggest accuracy >95% of the AI-enabled system. Analyses will then examine prevalence of claims on UPFs in comparison to non-UPFs. It is anticipated that there will be a difference in the types and prevalence of claims between child-targeted UPFs and non-UPFs as well as between breakfast cereals and snack bars.

CONCLUSION

This proof-of-concept study will show that AI systems are accurate in analysing the claims on child-targeted UPFs. Therefore, upon completion of this project it will be possible for these claims to be analysed on larger sample sizes, to better understand the scale of the issue in the Australian context. This information is critical for regulators to know in order to address problematic practices, therefore to enable children and parents to make healthier food choices.

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